

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE310	PROGRAMMING IN JAVA	3	0	2	4	
Course Weightage	ATT: 5 CA: 50 ETP: 45	Exam Category: X6: Mid Term Exam: Not Applicable – End Term Exam: Practical				
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT,ENTREPRENEURSHIP					

Course Outcomes :Through this course students should be able to

CO1 :: explain basic constructs of Java programming and apply them to solve the real-world problems

CO2 :: illustrate the Object-oriented programming principles to write efficient and reusable codes.

CO3 :: demonstrate the concept of inheritance to reuse and extend the features of existing class with access control

CO4 :: create the uses of abstract classes, interfaces and Lambda expressions.

CO5 :: manage errors and perform multithreaded I/O operations using exception handling and file streams.

CO6 :: utilize collections, generics, and JDBC for advanced Java applications

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	PROGRAMMING WITH JAVA	E. BALAGURUSAMY	MCGRAW HILL EDUCATION
	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	JAVA THE COMPLETE REFERENCE	HERBERT SCHILDT	MCGRAW HILL EDUCATION
R-2	INTRODUCTION TO JAVA PROGRAMMING	Y. DANIEL LIANG	PEARSON

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://docs.oracle.com/javase/8/	Updated and authentic documentation of latest features is available

Audio Visual Aids (AV)		
Sr No	(AV aids) (only if relevant to the course)	Salient Features
AV-1	Video lectures on basics constructs of Java Programming	http://freevideolectures.com/Course/2513/Java-Programming

AV-2	Basics elaborated with focus on professional certifications	https://www.youtube.com/watch?v=8EOhMRM-BfE&list=PLd3UqWTnYXOnujVvl3wiZfrFKUEg9jBeA
Software/Equipments/Databases		
Sr No	(S/E/D) (only if relevant to the course)	Salient Features
SW-1	IDE to develop simple to complex application development in convenient and faster way.	IntelliJ IDEA iDE
SW-2	For writing, compiling and testing basic java constructs	JDK 1.8, Command Prompt and Text Editor

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Introduction to Java(History and Features of Java, Java program structure, Writing simple Java class and main() method, Command-line arguments, Understanding JDK, JRE and JVM)	T-1 R-2	SW-1 SW-2 AV-2	Writing first java program, and using command line arguments.	Understanding the importance of Java programs	Lecture and discussion	Demonstration of Basic Java program along with JDK architecture
	Lecture 2	Introduction to Java(History and Features of Java, Java program structure, Writing simple Java class and main() method, Command-line arguments, Understanding JDK, JRE and JVM)	T-1 R-2	SW-1 SW-2 AV-2	Writing first java program, and using command line arguments.	Understanding the importance of Java programs	Lecture and discussion	Demonstration of Basic Java program along with JDK architecture

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Week 1	Lecture 3	Introduction to Java(History and Features of Java, Java program structure, Writing simple Java class and main() method, Command-line arguments, Understanding JDK, JRE and JVM)	T-1 R-2	SW-1 SW-2 AV-2	Writing first java program, and using command line arguments.	Understanding the importance of Java programs	Lecture and discussion	Demonstration of Basic Java program along with JDK architecture
Week 2	Lecture 4	Data In the Cart(Using primitive data types, Type conversion, Keywords, Identifiers, Variables, Access modifiers, static keyword, Wrapper class)	T-1 R-2	SW-1 SW-2	Working with Primitive data types, keywords, variables, and access modifiers.	Understanding the usage of different data types and Narrowing/ Widening and importance of static in Java and Autoboxing and Unboxing	Lecture cum demonstration and Discussion	Programs to demonstrate different data types, autoboxing and unboxing
	Lecture 5	Data In the Cart(Using primitive data types, Type conversion, Keywords, Identifiers, Variables, Access modifiers, static keyword, Wrapper class)	T-1 R-2	SW-1 SW-2	Working with Primitive data types, keywords, variables, and access modifiers.	Understanding the usage of different data types and Narrowing/ Widening and importance of static in Java and Autoboxing and Unboxing	Lecture cum demonstration and Discussion	Programs to demonstrate different data types, autoboxing and unboxing
	Lecture 6	Conditional Statements (Using if/else constructs and switch-case statements)	T-1 R-1	SW-1 SW-2 AV-1	Using if-else if-else ladder and switch statement	Understanding different ways to work with conditional statements	Lecture cum demonstration and Brainstorming	Program to display the grades of the student based on the marks
Week 3	Lecture 7	Loops(Working with for loop, while loop, do-while loop and for-each loop)	T-1 R-1	SW-1 SW-2 AV-1	Using different types of loops in Java	Working with for, while, do-while and for-each loop	Lecture cum demonstration and Discussion	Program to implement various loops in Java
	Lecture 8	Arrays and Enums (Fundamentals about Arrays, Multi-dimensional arrays, Array Access and Iterations, Using varargs, Enumerations)	T-1 R-1	RW-1	Creation and usage of one dimensional and two-dimensional arrays	Using onedimensional, twodimensional arrays and varargs	Lecture cum demonstration and Brainstorming	Program to find the grades of student with different number of academic tasks
	Lecture 9	OOP Concepts(Basics of class and objects, Writing constructors and methods, Overloading methods and constructors, this keyword, initializer blocks)	T-1 R-1	SW-1 SW-2 AV-1	Creating classes, initializing objects. Implementing method and constructor overloading. Using this keyword. Implementing Initializer blocks.	Understanding implementation of OOP concepts.	Lecture cum demonstration and Discussion	Programs implementing different oop concepts.

Week 4	Lecture 10	OOP Concepts(Basics of class and objects, Writing constructors and methods, Overloading methods and constructors, this keyword, initializer blocks)	T-1 R-1	SW-1 SW-2 AV-1	Creating classes, initializing objects. Implementing method and constructor overloading. Using this keyword. Implementing Initializer blocks.	Understanding implementation of OOP concepts.	Lecture cum demonstration and Discussion	Programs implementing different oop concepts.
	Lecture 11	OOP Concepts(Basics of class and objects, Writing constructors and methods, Overloading methods and constructors, this keyword, initializer blocks)	T-1 R-1	SW-1 SW-2 AV-1	Creating classes, initializing objects. Implementing method and constructor overloading. Using this keyword. Implementing Initializer blocks.	Understanding implementation of OOP concepts.	Lecture cum demonstration and Discussion	Programs implementing different oop concepts.
	Lecture 12	String Class(Constructors and methods of String and String Builder class)	T-1 R-1	RW-1 SW-2	Implementing the constructors and methods of String and String Builder class	Understanding different ways of creating Strings and using the functionalities. Using mutable String objects.	Lecture cum demonstration and Discussion	Program to implement the methods of String and String Builder
Week 5	Lecture 13	String Class(Constructors and methods of String and String Builder class)	T-1 R-1	RW-1 SW-2	Implementing the constructors and methods of String and String Builder class	Understanding different ways of creating Strings and using the functionalities. Using mutable String objects.	Lecture cum demonstration and Discussion	Program to implement the methods of String and String Builder
	Lecture 14	Inheritance and Polymorphism(Inheritance, Method overriding, super keyword, Object class and overriding toString() and equals() method, Using super and final keywords, instanceof operator)	T-1 R-1 R-2	RW-1 SW-1 AV-1 AV-2	Inheritance in java	Understanding the importance of IS-A relationship, Polymorphism and keywords in inheritance	Lecture cum demonstration and Discussion	Program to create subclasses and implement different keywords.
	Lecture 15	Inheritance and Polymorphism(Inheritance, Method overriding, super keyword, Object class and overriding toString() and equals() method, Using super and final keywords, instanceof operator)	T-1 R-1 R-2	RW-1 SW-1 AV-1 AV-2	Inheritance in java	Understanding the importance of IS-A relationship, Polymorphism and keywords in inheritance	Lecture cum demonstration and Discussion	Program to create subclasses and implement different keywords.

Week 6	Lecture 16	Inheritance and Polymorphism(Inheritance, Method overriding, super keyword, Object class and overriding toString() and equals() method, Using super and final keywords, instanceof operator)	T-1 R-1 R-2	RW-1 SW-1 AV-1 AV-2	Inheritance in java	Understanding the importance of IS-A relationship, Polymorphism and keywords in inheritance	Lecture cum demonstration and Discussion	Program to create subclasses and implement different keywords.
	Lecture 17				Test			
	Lecture 18	Abstract Class and Interface (Abstract method and abstract class, Interfaces, static and default methods.)	T-1 R-2	RW-1 SW-2 AV-1 AV-2	Implementing abstraction using abstract class and abstract methods. Implementing interfaces	Understanding the importance of abstract method, abstract class and interfaces	Lecture cum demonstration and Discussion	Program to implement abstract class and its inheritance
Week 7	Lecture 19	Operators(Working with Bit-wise, arithmetic, logical, and relational operators, Unary, assignment and Ternary operator, Operator precedence)	T-1 R-1	SW-1 SW-2	Using various operators and operator precedence	Understanding the working of Bit-wise operators, short-circuit operators and operator precedence	Lecture cum demonstration and Discussion	Program to demonstrate the working of different operators
SPILL OVER								
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
MID-TERM								
Week 8	Lecture 22	Nested Class and Lambda Expressions:(Nested Class, Understanding the importance of static and non-static nested classes, Local and Anonymous class, Functional Interface, Lambda expressions)	T-1 R-1		Implementing static, non-static nested classes, and local and anonymous classes.	Understanding the need of nested classes	Lecture cum demonstration and Brainstorming	
	Lecture 23	Nested Class and Lambda Expressions:(Nested Class, Understanding the importance of static and non-static nested classes, Local and Anonymous class, Functional Interface, Lambda expressions)	T-1 R-1		Implementing static, non-static nested classes, and local and anonymous classes.	Understanding the need of nested classes	Lecture cum demonstration and Brainstorming	

Week 8	Lecture 24	Nested Class and Lambda Expressions:(Nested Class, Understanding the importance of static and non-static nested classes, Local and Anonymous class, Functional Interface, Lambda expressions)	T-1 R-1		Implementing static, non-static nested classes, and local and anonymous classes.	Understanding the need of nested classes	Lecture cum demonstration and Brainstorming	
Week 9	Lecture 25	Utility Classes(Working with Dates)	T-1 R-1		Using Dates in Java Activity 1: Non-Proctored Code Based Test. Also used for short listing students for fast learner track	Using Date in Java applications	Lecture cum demonstration and Discussion	
	Lecture 26	Utility Classes(Working with Dates)	T-1 R-1		Using Dates in Java Activity 1: Non-Proctored Code Based Test. Also used for short listing students for fast learner track	Using Date in Java applications	Lecture cum demonstration and Discussion	
	Lecture 27	Exceptions and Assertions (Exception overview, Exception class hierarchy and exception types, Propagation of exceptions, Using try, catch and finally for exception handling, Usage of throw and throws, handling multiple exceptions using multi-catch, Autoclose resources with try-with resources statement, Creating custom exceptions, Testing invariants by using assertions)	T-1 R-1		Overview of Exception and its class hierarchy. Handling exceptions using try, catch and finally. Using multicatch, throw and throws. Activity 2: Proctored Code Based Test. Also used for short listing students for fast learner track.	understanding different types of Exceptions and propagation of exception	Lecture cum demonstration and Discussion	

Week 10	Lecture 28	Exceptions and Assertions (Exception overview, Exception class hierarchy and exception types, Propagation of exceptions, Using try, catch and finally for exception handling, Usage of throw and throws, handling multiple exceptions using multi-catch, Autoclose resources with try-with resources statement, Creating custom exceptions, Testing invariants by using assertions)	T-1 R-1		Overview of Exception and its class hierarchy. Handling exceptions using try, catch and finally. Using multicatch, throw and throws. Activity 2: Proctored Code Based Test. Also used for short listing students for fast learner track.	understanding different types of Exceptions and propagation of exception	Lecture cum demonstration and Discussion	
	Lecture 29	Exceptions and Assertions (Exception overview, Exception class hierarchy and exception types, Propagation of exceptions, Using try, catch and finally for exception handling, Usage of throw and throws, handling multiple exceptions using multi-catch, Autoclose resources with try-with resources statement, Creating custom exceptions, Testing invariants by using assertions)	T-1 R-1		Overview of Exception and its class hierarchy. Handling exceptions using try, catch and finally. Using multicatch, throw and throws. Activity 2: Proctored Code Based Test. Also used for short listing students for fast learner track.	understanding different types of Exceptions and propagation of exception	Lecture cum demonstration and Discussion	
	Lecture 30				Test - Code based 1			
Week 11	Lecture 31	I/O Fundamentals (Describing the basics of input and output in Java, Read and write data from various sources, Using streams to read and write files, Writing and read objects using serialization)	T-1 R-1		Use of Scanner, PrintWriter, FileInputStream, FileOutputStream, ObjectInputStream and ObjectOutputStream	Understanding how IO works in Java	Lecture cum demonstration and Discussion	
	Lecture 32	I/O Fundamentals (Describing the basics of input and output in Java, Read and write data from various sources, Using streams to read and write files, Writing and read objects using serialization)	T-1 R-1		Use of Scanner, PrintWriter, FileInputStream, FileOutputStream, ObjectInputStream and ObjectOutputStream	Understanding how IO works in Java	Lecture cum demonstration and Discussion	

Week 11	Lecture 33	Generics(Creating a custom generic class, Using the type inference diamond to create an object, Using bounded types and Wild Cards.)	T-1 R-1		Differentiating between Comparable and Comparator, use of TreeSet, HashMap, LinkedList Activity 3: Non-Proctored Code Based Test. Also used for short listing students for fast learner track.	Implementing generic classes and collections	Lecture cum demonstration and Discussion	
Week 12	Lecture 34	Multithreading (Threads) (Thread lifecycle.)	T-1 R-1		Thread lifecycle, Thread class, and Runnable interface	Understand thread states; create and manage threads using Thread and Runnable	Brief lecture, diagram explanation, live coding	
		Multithreading (Threads) (Thread class.)	T-1 R-1		Thread lifecycle, Thread class, and Runnable interface	Understand thread states; create and manage threads using Thread and Runnable	Brief lecture, diagram explanation, live coding	Java program demonstrating thread lifecycle and thread creation using Thread and Runnable
		Multithreading (Threads) (Runnable interface)	T-1 R-1		Thread lifecycle, Thread class, and Runnable interface	Understand thread states; create and manage threads using Thread and Runnable	Brief lecture, diagram explanation, live coding	
	Lecture 35	Multithreading (Threads) (Thread priorities)	T-1 R-1		Thread priorities, Synchronization, and Inter-thread communication	Understand thread priority levels; apply synchronization to avoid race conditions; implement inter-thread communication	Concept explanation with problem-based discussion, live coding	Java program demonstrating thread priorities, synchronized methods/blocks, and producer–consumer example
		Multithreading (Threads) (Synchronization)	T-1 R-1		Thread priorities, Synchronization, and Inter-thread communication	Understand thread priority levels; apply synchronization to avoid race conditions; implement inter-thread communication	Concept explanation with problem-based discussion, live coding	Java program demonstrating thread priorities, synchronized methods/blocks, and producer–consumer example

Week 12	Lecture 35	Multithreading (Threads) (inter-thread communication)	T-1 R-1		Thread priorities, Synchronization, and Inter-thread communication Activity 4: Proctored Code Based Test. Also used for short listing students for fast learner track.	Understand thread priority levels; apply synchronization to avoid race conditions; implement inter-thread communication	Concept explanation with problem-based discussion, live coding	
	Lecture 36	Collections(Creating a collection by using generics, Implementing an ArrayList, Implementing TreeSet using Comparable and Comparator interfaces, Implementing a HashMap, Implementing a Deque.)	T-1 R-1 R-2		Differentiating between Comparable and Comparator, use of TreeSet, HashMap, LinkedList	implementing generic classes and collections	Lecture cum demonstration and Discussion	
Week 13	Lecture 37	Collections(Creating a collection by using generics, Implementing an ArrayList, Implementing TreeSet using Comparable and Comparator interfaces, Implementing a HashMap, Implementing a Deque.)	T-1 R-1 R-2		Differentiating between Comparable and Comparator, use of TreeSet, HashMap, LinkedList	implementing generic classes and collections	Lecture cum demonstration and Discussion	
	Lecture 38				Test - Code based 2			
	Lecture 39	Java Database Programming (Introduction to JDBC)	T-1 R-1		Introduction to JDBC, JDBC Drivers	utilise collections, generics, and JDBC for advanced Java applications.	Lecture and discussion	Program to save data on Database
		Java Database Programming (JDBC Drivers)	T-1 R-1		Introduction to JDBC, JDBC Drivers	utilise collections, generics, and JDBC for advanced Java applications.	Lecture and discussion	Program to save data on Database
		Java Database Programming (CRUD operation Using JDBC)	T-1 R-1		Introduction to JDBC, JDBC Drivers	utilise collections, generics, and JDBC for advanced Java applications.	Lecture and discussion	Program to save data on Database
		Java Database Programming (Connecting to non-conventional Databases.)	T-1 R-1		Introduction to JDBC, JDBC Drivers	utilise collections, generics, and JDBC for advanced Java applications.	Lecture and discussion	Program to save data on Database
Week 14	Lecture 40	Java Database Programming (Introduction to JDBC)	T-1 R-1		Introduction to JDBC, JDBC Drivers	utilise collections, generics, and JDBC for advanced Java applications.	Lecture and discussion	Program to save data on Database

Week 14	Lecture 40	Java Database Programming (JDBC Drivers)	T-1 R-1		Introduction to JDBC, JDBC Drivers	utilise collections, generics, and JDBC for advanced Java applications.	Lecture and discussion	Program to save data on Database
		Java Database Programming (CRUD operation Using JDBC)	T-1 R-1		Introduction to JDBC, JDBC Drivers	utilise collections, generics, and JDBC for advanced Java applications.	Lecture and discussion	Program to save data on Database
		Java Database Programming (Connecting to non-conventional Databases.)	T-1 R-1		Introduction to JDBC, JDBC Drivers	utilise collections, generics, and JDBC for advanced Java applications.	Lecture and discussion	Program to save data on Database
		SPILL OVER						
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

Scheme for CA:

CA Category of this Course Code is:C020102 (Total 4 tasks, 2 compulsory and out of remaining 1 best out of 2 to be considered)

Component	Iscompulsory	Weightage (%)	Mapped CO(s)
Test - Code based 2	NO	30	CO4, CO5, CO6
Test - Code based 1	NO	30	CO1, CO2, CO3
Programming Practice	Yes	30	CO1, CO2, CO3, CO4, CO5, CO6
Test	Yes	40	CO1, CO2

Details of Academic Task(s)

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Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allotment / submission Week
Programming Practice	To evaluate progress of individual student based upon applicability of learned concepts	The students will be given approx. 90 coding problems and 90 MCQs on the selected third party platform. The questions will be framed with equal distribution from the complete syllabus. In order to qualify for programming practice marks, the student should solve at least 50% of the coding problems and 50% of MCQ questions (eligibility condition). The maximum marks out of 15 marks for which the student would be eligible for Programming Practice would be based on the Percentage of questions solved by the student Example- If a student solves 72 questions out of 90 questions (i.e. 80% of questions solved) then the student would be eligible for 80% of 15 marks which is 12 marks (round-up would be used in case of decimal values). The final marks for Programming Practice would be calculated by prorating the eligible marks for which the student is eligible (as explained in the above point) with the percentage of marks the student has scored in the proctored Coding Contests conducted as CA's along with the mandatory written test. (The final marks would be rounded up for the students). Example – If a student solves 72 questions out of 90 questions (i.e. 80% of questions solved) then the student would be eligible for 80% of 15 marks which is 12 marks (round-up would be used in case of decimal values). And if the student has scored 24 out of 30 in the other CA's i.e. 80% marks in CA, his Programming practice final marks would be 80% of 12 marks that he was eligible for which is 9.6 rounded up to 10 marks out of 15 for Programming Practice.	Individual	Online	15	1 / 13
Test - Code based 2	To evaluate basic programming skills of students.	Mix of MCQs (10 Marks) and Coding Problems (20 Marks)	Individual	Online	30	12 / 13
Test	To evaluate progress of individual student based upon applicability of learned concepts	This is a written Test which is madatory. The test will be of one hour duration . The test will cover the topics completed in week 5 and 6	Individual	Offline	30	5 / 6
Test - Code based 1	To evaluate basic programming skills of students.	Mix of MCQs (10 Marks) and Coding Problems (20 Marks)	Individual	Online	30	9 / 10

List of suggested topics for term paper[at least 15] (Student to spend about 15 hrs on any one specified term paper)

Sr. No.	Topic
1	submitted

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	Creating a Java Main Class	Program to implement a java class.		Understanding the main() method and how command-line arguments work in Java.
Practical 2	Managing Multiple Items	Program to demonstrate the use of list of items.		Understanding different ways to work with conditional and looping statements
Practical 3	Describing Objects and Classes	Program to demonstrate the instantiation of class and accessing the attributes using object of class.		Understanding the Object initialization and associating functionalities
Practical 4	Manipulating and Formatting the Data in Your Program	Program to demonstrate the uses of String and StringBuilder		Understanding different ways of creating Strings and using the functionalities
Practical 5	Using Inheritance	Program to demonstrate the inheritance and its importance using Swing Components.		Understanding the importance of IS-A relationship and Polymorphism
Practical 6	Overriding Methods, Polymorphism, and Static Classes	Program to implement polymorphism and using proper access control.		Understanding the importance of IS-A relationship and Polymorphism
Practical 7	Abstract and Nested Classes	Program to demonstrate the use of abstract class and nested class.		Identifying the need of interfaces and static, default and abstract methods
Practical 8	Abstract and Nested Classes	Program to demonstrate the use of abstract class and nested class.		Identifying the need of interfaces and static, default and abstract methods
Practical 9	Multithreading	Program to implement multithreading using Lambda Expressions.		Using java multithreading concept to make subtasks run in parallel.
Practical 10	Multithreading	Program to implement multithreading using Lambda Expressions.		Using java multithreading concept to make subtasks run in parallel.
Practical 11	Exception Handling	Program to demonstrate the use of all the keywords used for exception handling and need of assertion		Program to demonstrate the use of all the keywords used for exception handling and need of assertion
Practical 12	Exception Handling	Program to demonstrate the use of all the keywords used for exception handling and need of assertion		Program to demonstrate the use of all the keywords used for exception handling and need of assertion
Practical 13	Java IO	Program to implement read and write operation using console and File.		Understanding how IO works in Java
Practical 14	Java IO	Program to implement read and write operation using console and File.		Understanding how IO works in Java
	SPILL OVER			
Practical 15	Spill Over			

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